

Processando config 23, seed 1, teste: Rabbit
 xx
 Starting Rabbit: nb = 16777216
 Version: TestU01 1.2.3
 xx

 HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
 Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin

smultin_MultinomialBitsOver test:

 N = 2, n = 8388608, r = 0, s = 32, L = 38,
 Sparse = TRUE

 Number of bits = n = 8388608
 Number of cells = 2^L = 274877906944
 Expected number per cell = $1 / 32768$
 EColl = $n^2 / (2k)$ = 128
 Hashing = TRUE

 Collision test
 CollisionOver: density = $n / k = 1 / 32768.00$
 Expected number of collisions = $\mu = 128.00$

 Results of CollisionOver test:

POISSON approximation	:	
Expected number of collisions = $N \cdot \mu$	:	256.00
Observed number of collisions	:	289
p-value of test	:	0.02

 Total number of cells containing j balls

j = 0	:	549739036961
j = 1	:	16776638
j = 2	:	289
j = 3	:	0
j = 4	:	0
j = 5	:	0

 CPU time used : 00:00:04.49

Generator state:
 16777216 bits have been read.

 HOST = matheus-IdeaPad-3-15ALC6, Linux

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snpair_ClosePairsBitMatch test:

N = 1, n = 262144, r = 0, t = 2

Max of all bit distances : 17.00
p-value of test : 0.39

CPU time used : 00:00:00.05

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
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snpair_ClosePairsBitMatch test:

N = 1, n = 131072, r = 0, t = 4

Max of all bit distances : 8.00
p-value of test : 0.12

CPU time used : 00:00:00.07

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin

svaria_AppearanceSpacings test:

N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of $n = (K + Q)L = 15728640$ bits
Q = 524288 initialization blocks
K = 524288 blocks for the test
the blocks have L = 15 bits

Normal statistic : 0.81
p-value of test : 0.21

CPU time used : 00:00:00.00

Generator state:
67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
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scomp_LinearComp test:

N = 1, n = 18176, r = 0, s = 32

Number of degrees of freedom : 7
Chi2 statistic for size of jumps : 5.88
p-value of test : 0.55

Normal statistic for number of jumps : -51.77
p-value of test : 1 - eps1 *****

CPU time used : 00:00:00.04

Generator state:
67127040 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin

scomp_LempelZiv test:

N = 1, n = 16777216, r = 0, s = 32, k = 24

Normal statistic : -0.83
p-value of test : 0.80

CPU time used : 00:00:00.41

Generator state:

83904256 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin

sspectral_Fourier1 test:

N = 1, n = 1048576, r = 0, s = 32, k = 20

Normal statistic : 0.15
p-value of test : 0.44

CPU time used : 00:00:00.21

Generator state:

84952832 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin

sspectral_Fourier3 test:

N = 8065, n = 2048, r = 0, s = 32, k = 11

Kolmogorov-Smirnov+ statistic = D+ : 0.052
p-value of test : 0.06

Kolmogorov-Smirnov- statistic = D- : 5.72e-3
p-value of test : 0.96

Anderson-Darling statistic = A2 : 1.72
p-value of test : 0.13

CPU time used : 00:00:00.12

Generator state:

101728032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin

sstring_LongestHeadRun test:

N = 1, n = 60, r = 0, s = 32, L = 279616

Number of degrees of freedom : 3
Chi-square statistic : 2.41
p-value of test : 0.49

Global longest run of 1 : 22.00
p-value of test : 0.63

CPU time used : 00:00:00.05

Generator state:
118504992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin

sstring_PeriodsInStrings test:

N = 1, n = 524288, r = 0, s = 31

Number of degrees of freedom : 94
Chi-square statistic : 70.72
p-value of test : 0.97

CPU time used : 00:00:00.02

Generator state:
135282208 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin

sstring_HammingWeight test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Number of degrees of freedom : 22
Chi-square statistic : 25.05
p-value of test : 0.29

CPU time used : 00:00:00.00

Generator state:
152059424 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -1.92
p-value of test : 0.97

CPU time used : 00:00:00.00

Generator state:
168836640 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin

sstring_HammingCorr test:

N = 1, n = 262144, r = 0, s = 32, L = 64

Normal statistic : 0.081
p-value of test : 0.47

CPU time used : 00:00:00.00

Generator state:
185613856 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin

sstring_HammingCorr test:

N = 1, n = 131072, r = 0, s = 32, L = 128

Normal statistic : -0.21
p-value of test : 0.58

CPU time used : 00:00:00.00

Generator state:
202391072 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 171.23
p-value of test : 0.61

CPU time used : 00:00:00.00

Generator state:
219168288 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 291.22
p-value of test : 0.71

CPU time used : 00:00:00.00

Generator state:

235945504 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin

sstring_HammingIndep test:

N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 489
Chi-square statistic : 500.98
p-value of test : 0.34

CPU time used : 00:00:00.01

Generator state:

252722720 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin

sstring_AutoCor test:

N = 1, n = 16777153, r = 0, s = 32, d = 1

```
-----
Normal statistic          :   -0.25
p-value of test          :    0.60
-----
```

```
CPU time used            :   00:00:00.00
```

```
Generator state:
 269499904 bits have been read.
```

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*****
```

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HOST = matheus-IdeaPad-3-15ALC6, Linux
```

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ufile_CreateReadBin:   /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin
```

```
sstring_AutoCor test:
```

```
-----
      N =  1,   n = 16777154,   r =  0,   s = 32,   d = 2
-----
```

```
-----
Normal statistic          :    0.78
p-value of test          :    0.22
-----
```

```
CPU time used            :   00:00:00.01
```

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Generator state:
 286277088 bits have been read.
```

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*****
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```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:   /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin
```

```
sstring_Run test:
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```
-----
      N =  1,   n = 3355443,   r =  0,   s =  32
-----
```

```
-----
Total number of 1 runs:  3355446
```

```
Number of degrees of freedom      :   36
Chi2 statistic for number of runs :  35.89
p-value of test                   :    0.47
-----
```

```
-----
Total number of bits:  13423616
```

```
Normal statistic for number of bits :    0.36
p-value of test                   :    0.36
```

CPU time used : 00:00:00.04

Generator state:
299700704 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin

smarsa_MatrixRank test:

N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

Number of degrees of freedom : 3
Chi-square statistic : 3.56
p-value of test : 0.31

CPU time used : 00:00:00.07

Generator state:
316477920 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin

smarsa_MatrixRank test:

N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

Number of degrees of freedom : 2
Chi-square statistic : 1056.73
p-value of test : eps *****

CPU time used : 00:00:00.03

Generator state:
333169120 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin

swalk_RandomWalk1 test:

N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

Test on the values of the Statistic H

Number of degrees of freedom	:	42
ChiSquare statistic	:	30.06
p-value of test	:	0.92

Test on the values of the Statistic M

Number of degrees of freedom	:	42
ChiSquare statistic	:	31.55
p-value of test	:	0.88

Test on the values of the Statistic J

Number of degrees of freedom	:	64
ChiSquare statistic	:	53.18
p-value of test	:	0.83

Test on the values of the Statistic R

Number of degrees of freedom	:	36
ChiSquare statistic	:	31.34
p-value of test	:	0.69

Test on the values of the Statistic C

Number of degrees of freedom	:	22
ChiSquare statistic	:	17.18
p-value of test	:	0.75

CPU time used : 00:00:00.05

Generator state:
349946336 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin

swalk_RandomWalk1 test:

N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

Test on the values of the Statistic H

Number of degrees of freedom	:	94
ChiSquare statistic	:	88.02
p-value of test	:	0.65

Test on the values of the Statistic M

Number of degrees of freedom	:	94
ChiSquare statistic	:	90.88
p-value of test	:	0.57

Test on the values of the Statistic J

Number of degrees of freedom	:	512
ChiSquare statistic	:	523.51
p-value of test	:	0.35

Test on the values of the Statistic R

Number of degrees of freedom	:	90
ChiSquare statistic	:	83.46
p-value of test	:	0.67

Test on the values of the Statistic C

Number of degrees of freedom	:	50
ChiSquare statistic	:	33.17
p-value of test	:	0.97

CPU time used : 00:00:00.03

Generator state:

366723552 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed1.bin

swalk_RandomWalk1 test:

N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

Test on the values of the Statistic H

Number of degrees of freedom	:	130
ChiSquare statistic	:	128.48
p-value of test	:	0.52

Test on the values of the Statistic M

Number of degrees of freedom	:	131
ChiSquare statistic	:	124.50
p-value of test	:	0.64

Test on the values of the Statistic J

Number of degrees of freedom	:	161
ChiSquare statistic	:	148.87
p-value of test	:	0.74

Test on the values of the Statistic R

Number of degrees of freedom	:	130
ChiSquare statistic	:	107.11
p-value of test	:	0.93

Test on the values of the Statistic C

Number of degrees of freedom	:	89
ChiSquare statistic	:	93.03
p-value of test	:	0.36

CPU time used : 00:00:00.03

Generator state:

383500352 bits have been read.

===== Summary results of Rabbit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin
Number of bits:	16777216
Number of statistics:	39

Total CPU time: 00:00:05.98
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
-----	-----
5 LinearComp	1 - eps1
22 MatrixRank, 320 x 320	eps
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All other tests were passed